Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience developing **modular software systems**, **distributed services**, requirements-driven workflows, secure integrations, automated testing, and reliable cross-platform production applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant software components** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load.
- Achieved 99%+ reliability across 5K+ monthly jobs through **documented requirements-driven data workflows**, validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python system** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows reliably daily.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **modular software architecture** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a **distributed authentication architecture** spanning user, IoT-device, and role-based access scenarios, evaluating smart-contract verification, immutable identity records, and self-sovereign credentials to reduce reliance on centralized identity stores.
- Published peer-reviewed security research after analyzing **identity attack protections** against replay, phishing, man-in-the-middle, and credential theft, documenting access-control trade-offs and presenting technically defensible decentralized authentication designs at IEEE conferences.

Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, Bash
Software Design: Object-Oriented Design, Data Structures, Algorithms, System Design, Modular Architecture, Microservices, REST APIs, Interface Design
Systems & Platforms: Linux, Windows, macOS, Distributed Systems, Multithreading, Asynchronous Processing, Role-Based Access Control, WebSockets
Cloud & Data: AWS, Azure, GCP, PostgreSQL, MySQL, MongoDB, Redis, Azure SQL
Engineering Practices: Git, GitHub, Docker, Jenkins, CI/CD, Unit Testing, Integration Testing, PyTest, Mockito, Production Debugging, Requirements Analysis, Technical Documentation, Performance Optimization

Projects

Software Reuse Decision Framework - Design Concept

- Designed a **requirements and trade-off framework** concept comparing reuse, COTS/GOTS integration, and new development across interfaces, security, maintainability, licensing, deployment constraints, verification effort, operational risk, and lifecycle cost.

Mission Workflow Integration Service - Design Concept

- Designed a **modular integration service** concept with documented interfaces, adapters, validation, role-based authorization, failure isolation, audit logging, automated tests, deployment checks, and runbooks for maintainable system sustainment.

CPU-Scheduling Simulator | Live

- Implemented **FCFS, SJF, SRT, Round Robin, priority, and MLQ scheduling** with React controls, EJ-2 visualization, and dynamic turnaround and waiting-time calculations across 1K+ simulated process combinations and edge cases.